

UCS668: Edge AI and Robotics: Data Centre Vision

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Course Objective: This course will provide students with basic fundamental understanding and practical hands-on training of computer vision and deep learning models on data centre GPU servers.

Syllabus

Introduction: Introduction to Deep Learning, Formulating Computer Vision Problem Statements, Image Classification using CNN Architectures like VGG, Inception, ResNet(18/34/50/152). Working towards building Object detection and Segmentation pipelines, Moving towards various approaches to solve Medical Imaging Problems.

Introduction to Deep Learning: Introduction to Artificial Neural Networks (ANNs), Artificial Neurons, Layers, Perceptron, Multilayer Perceptron, Advanced Deep Neural Networks (DNNs), Batch Normalization, Hyperparameter tuning, Activation Functions, Metrics, Optimization, Regularization.

Applications of Computer Vision (Image Classification): Introduction to NVIDIA Frameworks: {Transfer Learning using Transfer Learning Toolkit (TLT), Mixed Precision, DALI}, Image Classification using Deep CNN Architecture like VGG, ResNet18/34/50, re-training on custom dataset.

Applications of Computer Vision (Object Detection & Segmentation): Introduction to Object Detection, Data Preprocessing, CNN Architecture like {SSD, YOLOv3}, Metrics, Loss Functions, re-training on custom dataset, Segmentation: FCN-ResNet, Unet, MaskRCNN, Metrics and Loss functions.

Advanced Vision and its Application in Medical Imaging: Introduction to Unsupervised Learning, Self-Supervised Learning, Medical Datasets, Generative Adversarial Networks.

Laboratory Work:

- Image Classification with RAPIDS-based Random Forest.
- Introduction to DL Frameworks: TLT, PyTorch, and Tensorflow (Keras).
- Binary Classification with Perceptron and Logistic Regression.
- {DLI Online Course: Getting Started with Deep Learning}
- {DLI Online Course: Deep Learning at Scale with Horovod}
- Training Classification Models with and without Mixed Precision and Multi-GPU on Open & Custom Datasets.
- Training Detection Models with and without Mixed Precision and Multi-GPU on Open & Custom Datasets.
- Training Segmentation Models with and without Mixed Precision and Multi-GPU on Open & Custom Datasets.
- {DLI Online Course: Getting Started with Image Segmentation}
- {DLI Online Course: Medical Image Classification Using the MedNIST Dataset}
- {DLI Online Course: Image Classification with TensorFlow: Radiomics — 1p19q Chromosome.

Course Learning Objectives (CLO)

After the completion of the course the student will be able to:

- 1 Understand the fundamental problems of computer vision.
2. Apply basic concepts of computer vision in various application domains which include robotics, healthcare, and graphics.
3. Analyze and evaluate the building and integration of computer vision algorithms and systems decomposition.
4. To implementation of various concepts in the key issues related to computer vision.

Text Books

1. Mitchell M., T., Machine Learning, McGraw Hill (1997) 1st Edition.
2. Alpaydin E., Introduction to Machine Learning, MIT Press (2014) 3rd Edition.
3. Vijayvargia Abhishek, Machine Learning with Python, BPB Publication (2018).
4. Computer Vision: Algorithms and Applications, R. Szeliski, Springer, 2011.
5. Computer Vision: A Modern Approach, D. Forsyth and J. Ponce, Prentice Hall, 2nd ed., 2011.

Reference Books:

1. Bishop M. C., Pattern Recognition and Machine Learning, Springer-Verlag (2011) 2nd Edition.
2. Introductory techniques for 3D computer vision, E. Trucco and A. Verri, Prentice Hall, 1998.
3. "Visualizing and Understanding Convolutional Networks" by Matthew D. Zeiler and Rob Fergus (2014)
4. "Convolutional Neural Networks for Visual Recognition" (Stanford course given by Fei-Fei Li, Andrej Karpathy, and Justin Johnson, 2016): <http://cs231n.github.io/>